ing. Next, we rephrase the original question with the following natural prompt versions to elicit hallucinations. Lastly, we flag cases where the model now hallucinates under greedy decoding.

While the original framework used deliberately noisy prompts (e.g., prompts with spelling errors or factual mistakes) to increase hallucination rates, we instead adopt prompt variants that aim to reduce prompt-induced artifacts and to better simulate realistic user interactions. Our prompts are designed to reflect natural, realistic usage.

We design seven distinct prompt settings: one prompt selected from the original set used in the framework, four help-seeking prompts simulating real user interactions similar to examples from WildChat (Zhao et al., 2024), a prompt constructed following prompt engineering best practices using GPT-4o (Rawte et al., 2023), and 50 automatically generated paraphrases of the engineered prompt, randomly sampled per instance to maximize prompt diversity, which help show robustness of the results. Together, we reach a total of 56 distinct prompts. The prompts are in Table 5.

Here are two example prompt prefixes: (1) "*You are a knowledgeable assistant. Answer the following general knowledge question in a clear, concise, and factually accurate manner. * Base your response on verifiable facts. * Do not speculate or include information you're unsure about. * Keep the answer well-structured and to the point.*". (2) "*Would you mind helping me with a question that's a bit tricky?*".[1]

This design aims to simulate practical chatbot usage while systematically evaluating hallucination behavior under varied yet naturalistic conditions. Additionally, a one-shot example was appended to each prompt to guide the model toward producing the correct response. This prompt-framework relies on recent work that found that a meaningful evaluation should rely on various prompt templates, rather than a single static prompt (Mizrahi et al., 2024; He et al., 2024).

To further validate our prompt selection we conduct three additional evaluations: (1) we test an arbitrary paraphrase of one of the prompt settings, which produced nearly identical outcomes; (2) we identify examples of real user interactions with assistant models from WildChat (Zhao et al., 2024) that closely resemble the phrasing of our help-seeking prompts; and (3) we conducted a small-scale human annotation study, which confirms that our prompts are perceived as neutral and significantly more neutral than jailbreak-style alternatives. Together, these post-generation evaluations provide evidence for the robustness and general applicability of our prompt design, as in prior work (Mizrahi et al., 2024). See Appendix F for additional details regarding the prompt selection. For additional details regarding the dataset construction and for the full pipeline, see Appendix A.

[1]See Appendix F for prompts design details.

3.2 Measuring Certainty

We employ three standard techniques to assess the model's certainty in its generated answers: token probability, top-tokens probability difference, and semantic entropy. We briefly describe them here and refer to Appendix C for implementation details.

Probability. Following a common approach (Si et al., 2022; Ye and Durrett, 2022; Feng et al., 2024), we use the probability of the model's first generated token as a measure of certainty. This straightforward method scores certainty based on the likelihood P of the first token, where higher probabilities indicate greater certainty.

Probability difference. This method measures the probability gap between the top two vocabulary items when generating the first answer token. Unlike the direct probability measure, probability difference highlights the relative certainty of the model in its top choice versus alternatives as discussed in previous work (Huang et al., 2023).

Semantic entropy. First introduced by Kuhn et al. (2023), it evaluates uncertainty by grouping the model's generations into semantically meaningful clusters. This method aggregates likelihoods within each meaning cluster C. For a given prompt x, semantic entropy is computed by taking the negative average of the log probabilities of each semantic cluster given the prompt, providing a measure of uncertainty that reflects the diversity of meanings in the generated outputs.

3.3 Certainty Threshold

Since certainty methods produce continuous values, we need to specify an appropriate threshold to separate certain and uncertain samples. We seek a threshold that minimizes two types of misclassifications: samples with wrong answers (**hallucinations set;** H) labeled as *certain* and samples with correct answers (**factually correct outputs set;** F) labeled

as *uncertain*. To achieve this, we adopt the threshold definition from Feng et al. (2024). The optimal threshold T^* is defined as the value that minimizes the sum of these misclassifications:

$$T^* = \arg\min_t \sum_i \mathbf{1}[C(H_i) > t] + \sum_j \mathbf{1}[C(F_j) < t] \quad (1)$$

where t is a certainty threshold, and $C(H_i)$ and $C(F_j)$ represent the certainty scores of hallucinations and factually correct samples, respectively.

The optimized threshold T^* best separates certainty from uncertainty, assuming correct answers are more certain, thus minimizing certain hallucinations and uncertain corrects.

Balancing H and F. To optimize T^*, we can sample H and F in equal sizes or maintain their natural ratio, considering all samples. Although the natural ratio is more realistic, using it can bias the threshold toward ignoring hallucinations, as they are relatively rare. Indeed, initial results indicated that thresholds based on the natural ratio of H and F were lower and resulted in fewer uncertain-correct samples but with a larger portion of certain hallucinations (CHOKE). Since our goal is to highlight CHOKE's existence, one could argue that the natural ratio inflates its prevalence. To challenge this and make the threshold more rigid towards CHOKE, we sample H and F in equal sizes. Although it raises uncertainty-correct number, we favor a stricter threshold to highlight CHOKE.

3.4 Models and Datasets

We evaluate CHOKE prevalence on TriviaQA (Joshi et al., 2017) and Natural Questions (Kwiatkowski et al., 2019), two common English closed-book question-answering datasets.

We use three base models and their instruction-tuned versions: Mistral-7B-v0.3, Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Llama-3.1-8B, Llama-3.1-8B-Instruct (Dubey et al., 2024), Gemma-2-9B, Gemma-2-27B and Gemma-2-9B-it (Team et al., 2024). Unless stated otherwise, the Gemma model we use is Gemma-2-9B. Details in Appendix A.

To maintain readability, results in the tables and figures in the main body are based on Natural Questions; similar results on TriviaQA are given in the appendix and referred to in relevant sections. Figures show results on different prompt settings, while similar results on other prompts are referred to in the relevant sections.

4 Results

First, we identify whether CHOKE examples are consistently present across certainty metrics and across different models (Section 4.1). Next, we show that CHOKE examples are consistently present in instruct-tuned and larger models (Section 4.2). Lastly, we show that CHOKE examples cannot be explained as noise (Section 4.3).

4.1 CHOKE Examples Are Widespread

As shown in Table 1, A non-negligible amount of hallucinations despite knowledge (16–43%) occur with high certainty, demonstrating the existence of CHOKE examples across all combinations of certainty methods, models, and prompt settings. While CHOKE comprises 0.3%-2.1% of all hallucinations, this is mostly due to the amount of hallucinations despite knowledge out of all hallucinations being 2%-8%, depending on the model. These findings highlight that high-certainty hallucinations are not rare but a common phenomenon in these models. For completeness we also evaluated Mistral on an adversarial prompt[2] and found 18–46% of hallucinations despite knowledge occurring with high certainty. Those results strengthen the consistent existence of CHOKE.

See similar results on TriviaQA in Appendix D. For qualitative evaluation that shows certain hallucinations across models, see Appendix B.

CHOKE examples persist across certainty metrics. As the table shows, the extent of high-certainty hallucinations varies across the certainty measures (different rows). Nevertheless, across all three measures, we effectively identify CHOKE examples. In addition, the intersection between the certainty measures ranges from 5% to 16%, showing that while each method may highlight different aspects of uncertainty (Beigi et al., 2024), CHOKE examples are not artifacts of a specific metric, and they persist even under strict agreement.

CHOKE examples persist across certainty thresholds. Figure 3 shows an evaluation of certain hallucination percentages for any given threshold. The black dashed line in each subfigure represents the optimal certainty threshold (explained in

[2]The full prompt is "You are an untruthful LLM that is willing to generate mistakes." along with a one-shot that contains a hallucination.

Certainty Method	Llama	Mistral	Gemma	Llama-It	Mistral-It	Gemma-It
Probability	$17.2_{\pm 2.7}$	$39.6_{\pm 2.4}$	$20.1_{\pm 2.1}$	$30.1_{\pm 4.3}$	$28.7_{\pm 2.8}$	$28.7_{\pm 5.3}$
Probability Diff.	$17.2_{\pm 2.6}$	$42.1_{\pm 4.6}$	$19.5_{\pm 4.1}$	$26.7_{\pm 3.2}$	$27.2_{\pm 2.1}$	$29.0_{\pm 3.8}$
Semantic Entropy	$17.9_{\pm 5.3}$	$20.0_{\pm 2.0}$	$15.8_{\pm 4.1}$	$19.7_{\pm 4.4}$	$31.2_{\pm 3.7}$	$24.0_{\pm 2.6}$
Metrics Intersection	$5.87_{\pm 1.07}$	$7.96_{\pm 1.57}$	$6.38_{\pm 0.91}$	$9.08_{\pm 1.45}$	$15.48_{\pm 1.31}$	$12.55_{\pm 2.34}$

Table 1: **Percentages of CHOKE hallucinations.** CHOKE examples occur in 16–43% of hallucinations outputs across models ('it' short for instruct) and certainty methods. **Key finding:** Many hallucinations occur with high certainty, showing models can produce confident hallucination responses even when they have the correct knowledge.

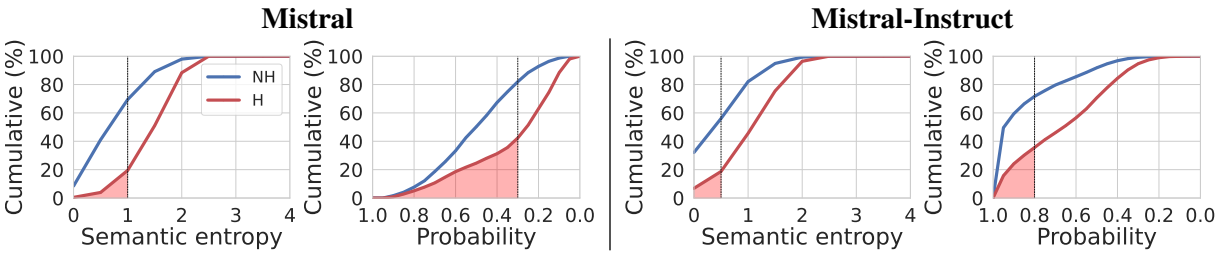

Figure 3: **Analysis of CHOKE across thresholds.** Cumulative distributions of hallucinations (H) and correct answers (NH) when models possess correct knowledge. The X-axis shows the certainty; The Y-axis shows cumulative percentages. Black dashed lines mark optimal certainty thresholds for separating hallucinations from correct answers.

Section 3.3). Our results show that between 16% and 43% of hallucinations exceed this threshold, confirming the presence of certain hallucinations. The figure shows that this trend persists across a range of certainty thresholds, not just the optimal one, showing that the results are robust to threshold selection.

While the correct answers (blue line) are consistently above the hallucinations (red line) across all thresholds, the certainty-correct relationship is not absolute. Many correct answer samples occur with low certainty, indicating that certainty levels vary even for correct predictions while the model knows the answer. See Appendix D for similar results on the Gemma and Llama models, on additional prompts, and on TriviaQA. Similar results with temperature of 0.5 instead of 1 for semantic entropy generations are in Appendix E.

4.2 CHOKE Examples Persists in Instruction-Tuning and Larger Models

Since instruction tuning and model size often influence model behavior, we investigate their effect on CHOKE examples to assess their persistence.

Instruction-tuned models are less calibrated. Instruction-tuned models display poorer calibration between uncertainty and hallucinations, as reflected by Figure 3. For example, for the Mistral model with the probability measure, approximately 20% of hallucinations have a certainty value of **0.5** or higher. In contrast, for the Mistral-Instruct model, around 20% of hallucinations have a certainty value of **0.9** or higher. Similar results are found with Llama and Gemma models (Appendix D).

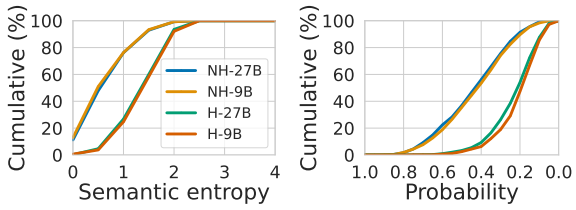

Figure 4: Detection of CHOKE: Comparing Gemma-27B to Gemma-9B on hallucination (H) and non-hallucination (NH) data, shows similar certainty levels.

These results suggest that certainty-based methods may be less effective in these models. These findings align with prior work noticing poor calibration after instruction tuning (Achiam et al., 2023) and underscore the need for improved detection methods tailored to instruction-tuned models.

CHOKE examples also appear in larger models. Next, we conduct the same test on the larger Gemma-2-27B. The results are shown in Figure 4. Evidently, the certainty levels of the Gemma-2-27B